

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks available |     |     |       |       |      |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        | (a) | (i)  |  | potassium / K <sup>+</sup><br>bromide / Br <sup>-</sup> do not accept bromine<br><br>calcium / Ca <sup>2+</sup><br>iodide / I <sup>-</sup> do not accept iodine<br><br>award (3) for all four ions correct<br>award (2) for any two correct<br>award (1) for any one correct                                                                                                                                                                                                                                                                                                         |                 |     | 3   | 3     |       | 3    |
|          |     | (ii) |  | BaCl <sub>2</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 | 1   |     | 1     |       |      |
|          | (b) |      |  | <div>Na<sup>+</sup>(aq) + NO<sub>3</sub><sup>-</sup>(aq) → AgCl(s) <input type="checkbox"/></div> <div>Ag<sup>+</sup>(aq) + NO<sub>3</sub><sup>-</sup>(aq) + Na<sup>+</sup>(aq) + Cl<sup>-</sup>(aq) → NaNO<sub>3</sub>(s) + AgCl(s) <input type="checkbox"/></div> <div>Ag<sup>+</sup>(aq) + NO<sub>3</sub><sup>-</sup>(aq) → AgNO<sub>3</sub>(s) <input type="checkbox"/></div> <div>Ag<sup>+</sup>(aq) + Cl<sup>-</sup>(aq) → AgCl(s) <input checked="" type="checkbox"/></div> <div>Ag<sup>+</sup>(aq) + Cl<sup>-</sup>(aq) → NaNO<sub>3</sub>(s) <input type="checkbox"/></div> |                 | 1   |     | 1     |       |      |